TABLEAU PORTFOLIO PROJECT

GROUP B: LOD EXPRESSIONS -- GEOSPATIAL PROFIT ANALYSIS

LOD Expressions & Geospatial Profit Analysis

INCLUDE, FIXED, and EXCLUDE LOD calculations on US choropleth and bubble maps

Tableau | LOD Expressions | FIXED | INCLUDE | EXCLUDE | Choropleth | Bubble Map | Geospatial

Project Summary

Demonstrates all three Tableau Level of Detail (LOD) expression types -- INCLUDE, FIXED, and EXCLUDE -- applied to a US geospatial profit dataset. Side-by-side choropleth maps compare results with and without LOD calculations, a bubble map shows city-level profit ratios relative to state totals, and a combined view overlays both LOD INCLUDE choropleth and LOD EXCLUDE bubbles on a single US map.

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1 PROJECT OVERVIEW

Workbook:	LOD Expressions -- Geospatial Profit Analysis
Views:	4 (LOD Include choropleth, LOD Fixed x2, LOD Fixed bubble map, Combined)
Data Source:	US retail dataset -- orders, profit, city/state geography
LOD Types Used:	INCLUDE (city-level within state), FIXED (country-level), EXCLUDE (state dimension)
Chart Types:	Dual choropleth US maps, proportional symbol bubble map, combined overlay map
Map Provider:	Mapbox / OpenStreetMap via Tableau
Business Use Case:	Profit analysis at multiple geographic granularities simultaneously

2 UNDERSTANDING LOD EXPRESSIONS

LOD (Level of Detail) expressions let analysts compute aggregations at a granularity DIFFERENT from the view level. The three types differ in how they relate to the view's own level of detail:

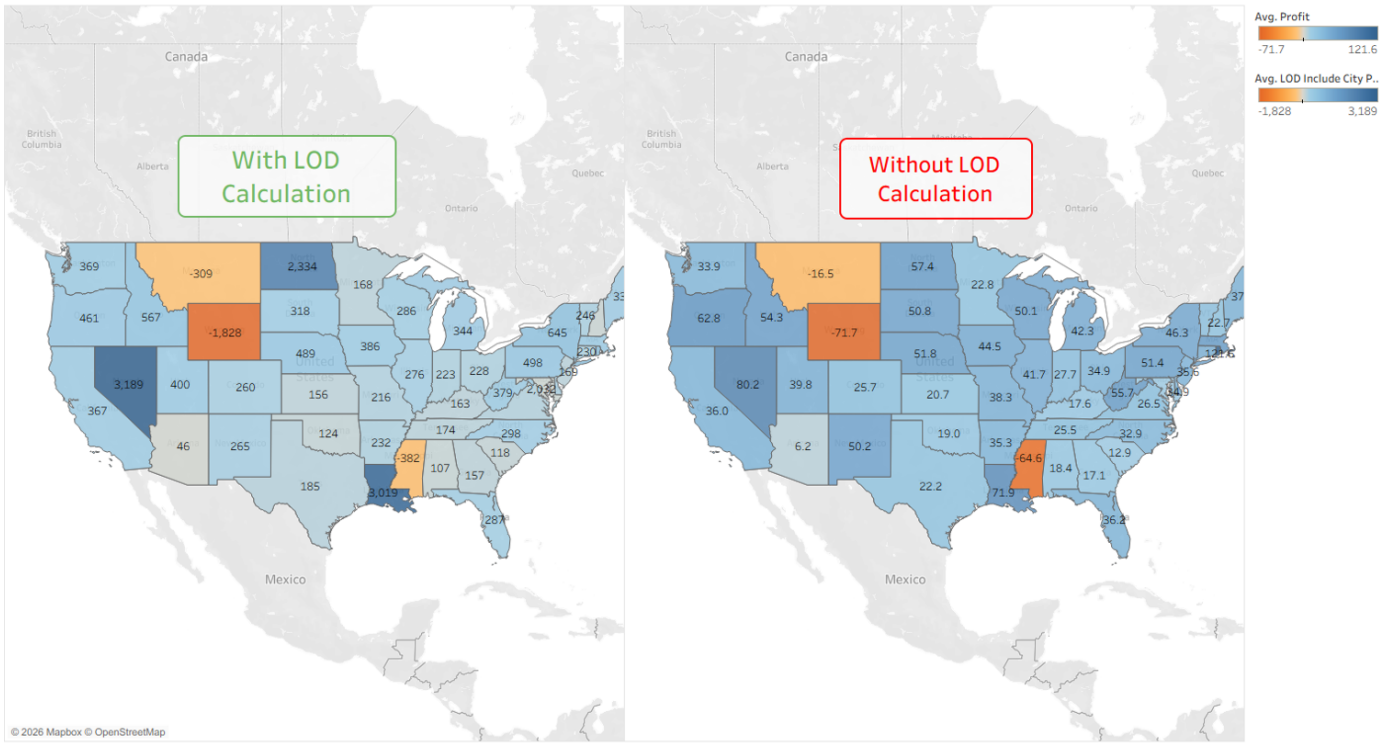
- INCLUDE: Adds dimensions to the aggregation. {INCLUDE [City] : SUM([Profit])} computes city-level profit even when the view is at state level -- producing more granular values than the view.
- FIXED: Computes at a fixed dimension regardless of view filters or dimensions. {FIXED [Country] : SUM([Profit])} gives total country profit on every row -- useful as a denominator for ratio calculations.
- EXCLUDE: Removes dimensions from the aggregation. {EXCLUDE [State] : SUM([Profit])} computes profit without the state dimension -- useful for comparing a city against the national average rather than state average.

Key insight: The side-by-side maps in this workbook make the LOD impact visually obvious. The "With LOD" map shows dramatically wider value ranges (e.g., -1,828 to 3,189) vs. the "Without LOD" map (-71.7 to 121.6) -- demonstrating that LOD expressions preserve sub-level variation that simple aggregates obscure.

3 VIEW 1 -- LOD INCLUDE: CITY PROFIT WITHIN STATE

Side-by-side US choropleth maps comparing Avg. LOD Include City Profit (left, -1,828 to 3,189 scale) vs. Avg. Profit without LOD (right, -71.7 to 121.6 scale). The LOD version reveals extreme profit variance at city level that plain state-level averaging masks. Colorado shows -1,828 (LOD) vs. its mild state average without LOD.

LOD Include



Map based on Longitude (generated) and Longitude (generated) and Latitude (generated). Details are shown for Country and State. For pane Longitude (generated): Color shows average of LOD Include City Profit. The marks are labeled by average of LOD Include City Profit. For pane Longitude (generated) (2): Color shows average of Profit. The marks are labeled by average of Profit.

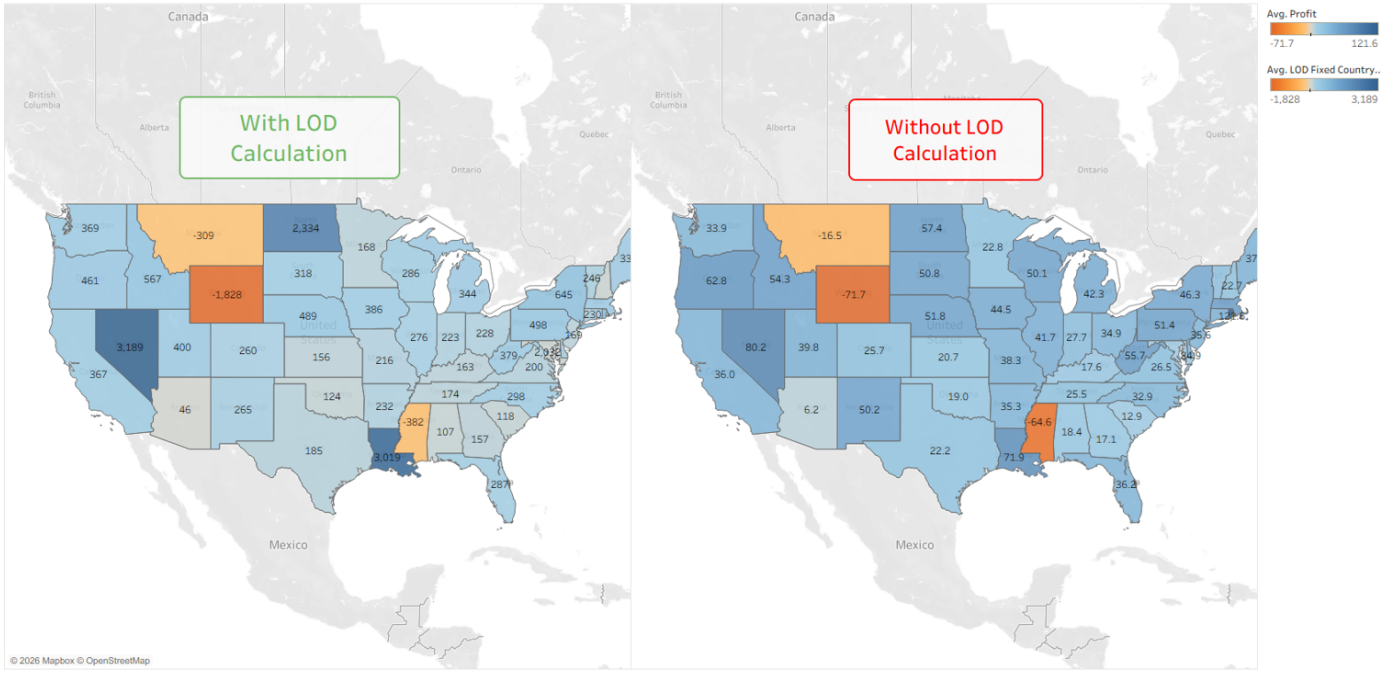
LOD INCLUDE -- Dual US choropleth: "With LOD Calculation" (city-level, range -1828 to 3189) vs "Without LOD" (state avg, range -71.7 to 121.6)

Tableau formula: {INCLUDE [City] : SUM([Profit])} -- forces city granularity even on a state-level map, revealing that some states contain both very profitable and very unprofitable cities that average out to a misleading state total.

4 VIEW 2 -- LOD FIXED: COUNTRY-LEVEL PROFIT COMPARISON

Similar side-by-side dual map using FIXED LOD. The FIXED expression anchors the calculation at country level regardless of any state or city filters applied in the view, making it useful as an absolute benchmark. The same -1,828 to 3,189 / -71.7 to 121.6 scales confirm the LOD impact is consistent across expression types.

LOD Fixed (1)



Map based on Longitude (generated) and Longitude (generated) and Latitude (generated). Details are shown for Country and State. For pane Longitude (generated): Color shows average of LOD Fixed Country Profit. The marks are labeled by average of LOD Fixed Country Profit. For pane Longitude (generated) (2): Color shows average of Profit. The marks are labeled by average of Profit.

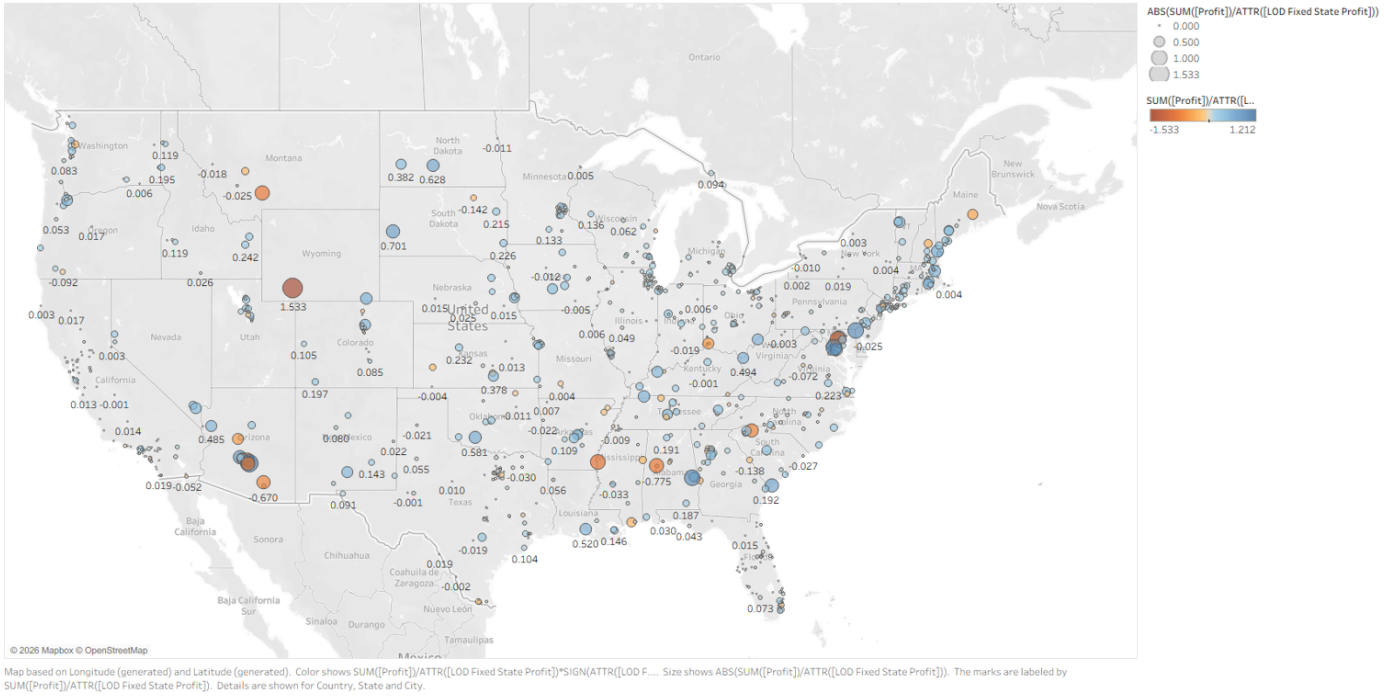
LOD FIXED (1) -- Dual US choropleth: "With LOD Calculation" (LOD Fixed Country Profit) vs "Without LOD" (plain Avg. Profit)

Tableau formula: {FIXED [Country] : SUM([Profit])} -- returns total US profit on every row. Used as denominator for ratio calculations: SUM([Profit]) / ATTR({FIXED [Country] : SUM([Profit])}) gives each state's share of national profit.

5 VIEW 3 -- LOD FIXED BUBBLE MAP: CITY PROFIT RATIO TO STATE TOTAL

City-level proportional symbol (bubble) map. Bubble SIZE encodes ABS(SUM([Profit]) / ATTR({FIXED [State] : SUM([Profit])})) -- the absolute ratio of city profit to state profit total. Bubble COLOR encodes the signed ratio (orange = negative ratio, blue = positive), revealing which cities are profit drains vs. profit drivers within their states.

LOD Fixed (2)



Map based on Longitude (generated) and Latitude (generated). Color shows SUM([Profit])/ATTR([LOD Fixed State Profit])*SIGN(ATTR([LOD F.... Size shows ABS(SUM([Profit])/ATTR([LOD Fixed State Profit])). The marks are labeled by SUM([Profit])/ATTR([LOD Fixed State Profit]). Details are shown for Country, State and City.

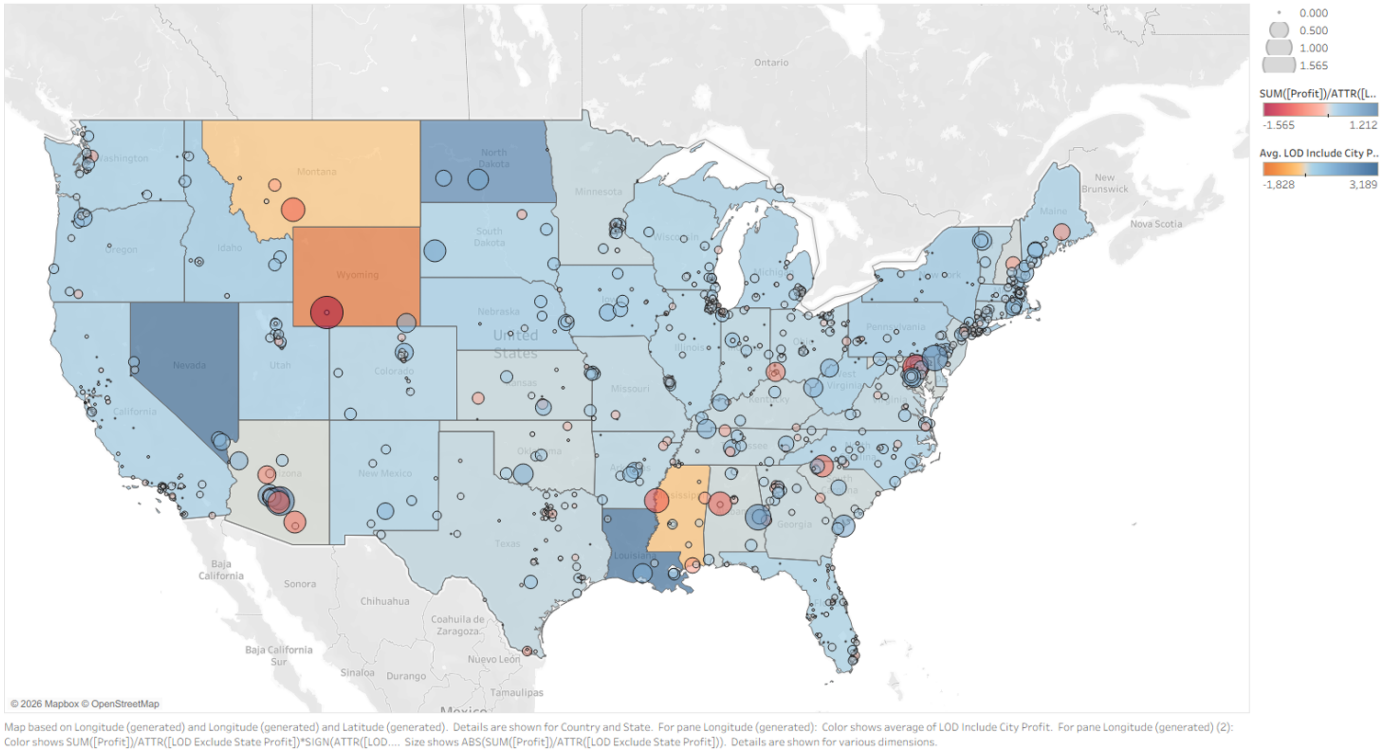
LOD FIXED (2) -- City bubble map. Size = ABS(city profit / state profit). Color = signed ratio (-1.533 orange to 1.212 blue). Large orange = city is pulling state profit negative.

- Wyoming shows ratio 1.533 -- single large orange bubble -- the state's profit is dominated by one city pulling it deeply negative.
- East Coast corridor shows many small blue bubbles -- diverse cities each contributing modest positive profit ratios.
- Arizona (-0.670) shows a notable negative outlier city within an otherwise positive state.

6 VIEW 4 -- COMBINED VISUALIZATION: LOD INCLUDE + EXCLUDE OVERLAY

The most advanced view overlays two LOD types on a single US map: the background choropleth uses LOD INCLUDE (city profit coloring state fills), while the foreground bubbles use LOD EXCLUDE (profit excluding the state dimension, comparing cities to national rather than state average). This dual-layer map supports both macro (state) and micro (city vs. national) analysis simultaneously.

Combined Visualization



Map based on Longitude (generated) and Longitude (generated) and Latitude (generated). Details are shown for Country and State. For pane Longitude (generated): Color shows average of LOD Include City Profit. For pane Longitude (generated) (2): Color shows SUM([Profit])/ATTR([LOD Exclude State Profit])*SIGN(ATTR([LOD.... Size shows ABS(SUM([Profit])/ATTR([LOD Exclude State Profit])). Details are shown for various dimensions.

Combined Visualization -- LOD INCLUDE choropleth background + LOD EXCLUDE proportional symbol overlay. Two LOD types on one map.

> *Tableau technique: Dual-axis map with two longitude pills on Columns shelf. Each axis independently controls mark type, color, and size. LOD INCLUDE drives the filled polygon layer; LOD EXCLUDE drives the bubble layer. Axes are synchronized to align geography.*

7 SKILLS DEMONSTRATED

- All 3 LOD expression types: INCLUDE, FIXED, EXCLUDE -- each applied to a distinct analytical question.
- Dual choropleth comparison layout: Side-by-side maps on same worksheet using dual Longitude pills.
- Proportional symbol map: Bubble size + color independently encoding two dimensions of a ratio.
- Combined dual-layer map: Choropleth polygons + bubble symbols on a single view via dual-axis technique.
- Mapbox satellite and street tile integration via Tableau map layers.
- Diverging color scales: Orange-to-blue encoding negative-to-positive profit ratios.